

Computer Vision Practical Report:

IMAGE TO BIGGER & SMALLER SIZES

1. AIM

To read an input image using OpenCV, resize the image to bigger and smaller sizes, display the original, bigger, and smaller images using Matplotlib, and save the resized images to disk.

2. PROCEDURE

- Load the input image using `cv2.imread('E1-IMG.png')`.
- Resize the image to a bigger size using `cv2.resize()`.
- Resize the image to a smaller size using `cv2.resize()`.
- Create a Matplotlib figure to display all three images.
- Display the original, bigger, and smaller images using `plt.subplot()`.
- Set titles as '**Original Image**', '**Bigger Image**', and '**Smaller Image**'.
- Disable axes using `plt.axis('off')`.
- Display the images using `plt.show()`.
- Save the resized images as **bigger_image.png** and **smaller_image.png** using `cv2.imwrite()`.

3. PROGRAM CODE

```
import cv2
import matplotlib.pyplot as plt

img_bgr = cv2.imread('E8-IMG.png')
img_rgb = cv2.cvtColor(img_bgr, cv2.COLOR_BGR2RGB)

height, width = img_rgb.shape[:2]

bigger_img = cv2.resize(img_rgb, (width * 2, height * 2), interpolation=cv2.INTER_CUBIC)
smaller_img = cv2.resize(img_rgb, (width // 2, height // 2), interpolation=cv2.INTER_AREA)

plt.figure(figsize=(12, 6))
```

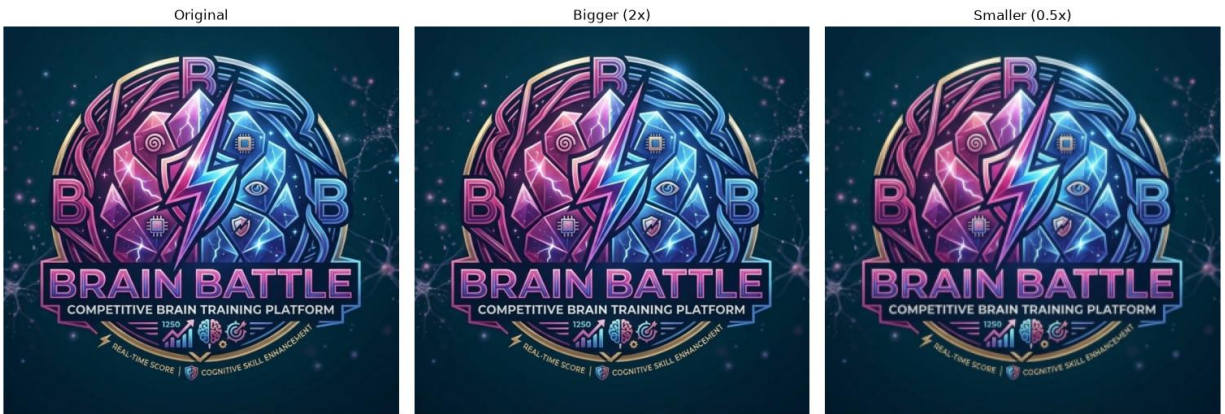
```
plt.subplot(1, 3, 1)
plt.imshow(img_rgb)
plt.title('Original')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(bigger_img)
plt.title('Bigger (2x)')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(smaller_img)
plt.title('Smaller (0.5x)')
plt.axis('off')

plt.tight_layout()
plt.show()
```

4.OUTPUT



5. RESULT

The Python script executed successfully. The input image 'E8-IMG.png' was loaded.